

UNIT-4 SOIL POLLUTION

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* Soil pollution :-

Additional of substances to the soil which adversely affect the physical, chemical and biological properties of the soil reduce its productivity.

Contamination of soil by human and natural activities which may cause harmful effect on living organism.

* Soil pollutant

Sources of soil pollutant substances which deteriorates the quality, texture and mineral content in soil and disturb the biological balance of organisms is known as soil pollutants.

following are the soil pollutant:-

- (i) persistent Toxic Compounds
- (ii) Chemicals
- (iii) Salts
- (iv) Radio Active materials
- (v) Disease causing agents
- (vi) Solid waste/ Garbage

(iii) Dumping Site and sewage tanks.

* Source of Soil pollution :-

① Industrial Activities

Disposal of Industrial waste is the major problem for Soil pollution.

Industrial pollutants are mainly discharged from various origin such as paper mills, Chemical fertilizers, oil refineries, Sugar factories, textile Industries, pesticides, Coal and Mining Industries etc.

② Agricultural Activities

• Modern Agricultural practices pollute the Soil to a large extent with increase in agro-technology huge quantities of fertilizers, pesticides, herbicides and weedicides are added to increase the crop yield. However use of these have long term effect on living organism.

• Apart from these, Soil erosion containing mostly Inorganic Chemicals are reported to cause Soil pollution.

⑬ Urban waste :—

- Urban waste contains both commercial and domestic waste consisting of dried sludge and sewage. All the urban solid waste are commonly refers as refuse.
- Waste are commonly of garbage and materials like plastic, glasses, metallic cans, fibre, paper, rubber, abandoned vehicle and other discarded manufactured products.
- Urban domestic waste though disposed off separately from industrial waste can still be dangerous.

⑭ Mining :—

Surface mining destroys landscape, forest and wildlife habitats at the site of the mine when trees, plant and topsoils are cleared from the mining area.

This in turn lead to soil erosion and disturbance of agricultural land. When rain washes the loosened top soil into stream sediments pollute waterways.

⑤ Deforestation :-

- Forest and grassland are excellent binding materials that ^{keep} ~~give~~ the soil intact and healthy.
- It results in floods and thereby causing soil erosion which degrades the quality of soil.

⑥ Radioactive pollutants :-

Radioactive substances resulting from explosion of nuclear testing laboratories and industries give rise to nuclear dust radioactive waste, penetrates the soil and accumulate giving rise to soil pollution.

⑦ Biological Agent :-

Soil gets a large amount of human animals and bird excreta which constitute major sources of land pollution by biological agents.

★ Types of Soil pollution

- Agricultural Soil ~~pollution~~ pollution
- Industrial effluents and Solid waste
- pollution due to Urban activities

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* Effect of Soil pollution

- ① Agriculture :-
 - ① Reduced Soil fertility
 - ② Reduced Nitrogen fixation
 - ③ Increased Soil Erosion
 - ④ Loss of Soil and Nutrient
 - ⑤ Reduced crop yield.
 - ⑥
- ② Health
 - ① Dangerous chemical entering under-ground water.
 - ② Biomagnification.
 - ③ Release of pollutant gases.
 - ④ Release of radioactive rays - causing health problem.

* Control of Soil pollution :-

- ① Afforestation or Reforestation and pressing preserving top Soil.
- ② Reducing chemical fertilizer and pesticide.
- ③ Use of biopesticides and biofertilizers.
- ④ Organic farming i.e use of biofertilizers.
- ⑤ Should follow four "Rs" :- Refuse Reduce Recuse and Reuse.

④ Industrial waste should be properly treated by physical, chemical and biological methods.

⑤ Areas usefit for Agriculture should be use dumping discarded materials ad quorbage.